

Product Backlog: AEGIS

This document provides the user stories that define the functional specifications of the platform. These stories form the foundation of the product backlog and will be continuously fine tuned through clear acceptance criteria for sprint planning.

User Personas

- **Recruiter:** The Admin or technical team member who creates, manages, and reviews candidate assessments.
- **Candidate:** The job applicant completing the technical evaluation.

Epic 1: Question & Assessment Management (Recruiter)

US1.1: AI-Resistant Question

User Story:

As a recruiter, I want to generate modified versions of coding questions that are resistant to AI-generated answers, so I can evaluate a candidate's true problem-solving abilities.

Acceptance Criteria:

- **Criteria 1.1.1: Transformation Rule**
 - **Given:** An existing coding question.
 - **When:** The recruiter requests an AI-resistant modification.
 - **Then:** The system must transform the text/constraints while preserving the original programming concept and difficulty level.
- **Criteria 1.1.2: Review Interface**
 - **Given:** A generated question.
 - **When:** Displayed to the recruiter.
 - **Then:** The platform must provide options to edit, regenerate, accept, or discard the question.
- **Criteria 1.1.3: Automated Verification Boundary:**
 - **Given:** A generated question variation.
 - **When:** The AI engine creates it.
 - **Then:** A background compilation script must automatically test the known solution against the new constraints to verify it resolves successfully without any errors or infinite loops before showing it to a recruiter.

US1.2: Question Bank Management

User Story:

As a recruiter, I want to manage and search coding questions in a question bank, so I can select suitable questions when creating assessments.

Acceptance Criteria:

- **Criteria 1.2.1: Search & Multi-Filter**
 - **Given:** The Question Bank dashboard.
 - **When:** A recruiter enters text in the search bar or applies a filter for difficulty, language, category, or Estimated Completion Time.
 - **Then:** The question list must update instantly to display only matching records.
- **Criteria 1.2.2: CRUD Operations**
 - **Given:** The Question Bank dashboard.
 - **When:** A recruiter to create, read, update, and delete any question.
 - **Then:** The system must execute the requested operation and update the repository state in the database.
- **Criteria 1.2.3: Categorization**
 - **Given:** A question details page.
 - **When:** A recruiter adds custom text strings into the tags field and saves.
 - **Then:** The system must associate those custom strings with a question.

US1.3: Assessment Creation & Assignment

User Story:

As a recruiter, I want to create and assign assessments by selecting questions and adjusting assessment settings, so candidates can complete structured technical assessments.

Acceptance Criteria:

- **Criteria 1.3.1.a: Successful Validation**
 - **Given:** A recruiter is on the assessment creation form.
 - **When:** They fill out all mandatory fields (Title, Description, Time Limit, Availability, Allowed Languages, Max Attempts) and click "Save".
 - **Then:** The system must successfully save the assessment draft.

- **Criteria 1.3.1.b: Failed Validation Error**
 - **Given:** A recruiter is on the assessment creation form.
 - **When:** They leave a mandatory field empty and click “Save”.
 - **Then:** The system must block the save action and display an error message next to the missing field(s).
- **Criteria 1.3.2: Preview Engine Mode**
 - **Given:** A set up assessment draft.
 - **When:** A recruiter clicks the “Preview” button
 - **Then:** The platform must render the complete assessment layout exactly as the candidate would experience it while keeping the countdown timer paused.
- **Criteria 1.3.3: Candidate Status Tracking**
 - **Given:** The recruiter is viewing the assessment tracking dashboard.
 - **When:** A candidate’s progress changes.
 - **Then:** The roster view must dynamically update that candidate’s status badge to match (Not Started, In Progress, or Completed).

US1.4: Assessment Duplications

User Story:

As a recruiter, I want to manage existing

Acceptance Criteria:

- **Criteria 1.4.1: Assessment Duplication**
 - **Given:** An existing assessment template.
 - **When:** The recruiter selects the “Duplicate” command.
 - **Then:** The system must generate a new, separate draft populated with identical questions, settings and metadata.
- **Criteria 1.4.2: Assessment Archiving**
 - **Given:** An active or completed assessment.
 - **When:** The recruiter clicks the “Archive” button.
 - **Then:** The system must hide the test from the main dashboard whilst keeping all of the associated candidate submission history in the database.

Epic 2: Candidate Evaluation & AI Insights (Recruiter)

US2.1: Candidate Performance Summaries

User Story:

As a recruiter, I want to view candidate performance summaries, so I can compare applicants and make informed hiring decisions.

Acceptance Criteria:

- **Criteria 2.1.1: Metric Sorting and Filtering**
 - **Given:** The candidate performance list view.
 - **When:** The recruiter clicks on the Overall Score, Time Taken , or AI Severity Flag column headers.
 - **Then:** The rows must re-order dynamically in ascending order or descending order.
- **Criteria 2.1.2: Data Export Extraction**
 - **Given:** A list of candidate results.
 - **When:** The recruiter clicks on the “Export” button.
 - **Then:** The system must compile all performance fields and trigger a local download of a .csv file.

US2.2: Submission Review & AI Integrity Indicators

User Story:

As a recruiter, I want to review candidate submissions and AI usage indicators, so I can determine whether a candidate displayed genuine technical ability.

Acceptance Criteria:

- **Criteria 2.2.1: Execution Review Panel**
 - **Given:** The recruiter is reviewing an individual candidate profile.
 - **When:** The review page loads.
 - **Then:** The UI must display the raw submitted source code side-by-side with a summary of passed or failed test cases and raw compilation or error messages.

- **Criteria 2.2.2: AI Monitoring Readout**
 - **Given:** A processed candidate submission panel.
 - **When:** The background processing completes.
 - **Then:** The system must display an AI Probability Flag score (Low, Medium, High) and list the matching monitoring indicators that triggered it.
- **Criteria 2.2.3: Automated Disqualification Guardrail**
 - **Given:** A candidate submission receives a High severity AI usage flag
 - **When:** The background processing completes.
 - **Then:** The system must add the flag to the profile but must not change the application workflow status to “Rejected” without manual recruiter intervention.

Epic 3: Technical Assessment Environment (Candidate)

US3.1: Code Editor & Execution Engine

User Story:

As a candidate, I want to answer assessment questions using a code editor and test my solutions, so I can complete the assessment and verify my implementation.

Acceptance Criteria:

- **Criteria 3.1.1: IDE Presentation**
 - **Given:** The candidate opens an active assessment question.
 - **When:** The code editor activates.
 - **Then:** It must apply syntax highlighting, tab spacing, and bracket auto-closing adjusted to the selected programming language.
- **Criteria 3.1.2: On-Demand Code Execution**
 - **Given:** A candidate has written code in the text canvas.
 - **When:** They click the “Run Code” button.
 - **Then:** Then the backend engine must execute the script against public test suites and display the success or failure outputs within 3 seconds.
- **Criteria 3.1.3: Enforced Assessment Locking**
 - **Given:** A candidate is completing an assessment.
 - **When:** A candidate manually submits the or the countdown timer reaches 00:00.

- **Then:** The code canvas must immediately switch to a read-only state, block all typing inputs, and write the final save state to the master database.

US3.2: Automated Progress Saving

User Story:

As a candidate, I want my assessment progress and answers to be saved automatically or manually, so I don't lose my work during an assessment.

Acceptance Criteria:

- **Criteria 3.2.1: Periodic Background Auto-Save**
 - **Given:** A candidate is actively typing in the workspace.
 - **When:** After 30 seconds elapses without a system failure.
 - **Then:** The platform must silently push the current code string to cloud storage and refresh the UI label to read: "Last saved at [Timestamp]".
- **Criteria 3.2.2: State Recovery on Re-Entry**
 - **Given:** A candidate loses their browser connection or crashes mid-test.
 - **When:** They re-open their unique assessment access link within the remaining exam window.
 - **Then:** The workspace must restore the exact code draft strings, question navigation state, and exact remaining time block down to the second.

US3.3: Results Transparency & Policy Guardrails

User Story:

As a candidate, I want my assessment platform to clearly explain the rules and expectations, so I can understand my performance and identify areas for development.

Acceptance Criteria:

- **Criteria 3.3.1: Pre-Assessment Rule Checkpoint**
 - **Given:** A candidate accesses the initial exam entry URL.
 - **When:** The welcome splash page loads.
 - **Then:** The system must disable the "Start Assessment" button until the candidate clicks the acknowledgement box for Duration, Scope, and AI Integrity Policies.
- **Criteria 3.2.2: Abstracted AI Flag Feedback**
 - **Given:** The recruiter has set up candidate feedback access to "Enabled".
 - **When:** A candidate reviews a flagged exam submission.

- **Then:** The UI must present high-level, human-readable explanations (e.g., structural text pasting patterns) while completely omitting internal heuristic weights, detection, parameters, or source logic.